PHYSICS MODULE 6

Electromagnetism

PHYSICS MODULE 6: ELECTROMAGNETISM

CHARGED PARTICLES, CONDUCTORS AND ELECTRIC AND MAGNETIC FIELDS

Inquiry question: What happens to stationary and moving charged particles when they interact with an electric or magnetic field?

- investigate and quantitatively derive and analyse the interaction between charged particles and uniform electric fields, including: (ACSPH083)
  - electric field between parallel charged plates $E = \frac{V}{d}$
  - acceleration of charged particles by the electric field $\vec{F}_{\text{net}} = m\vec{a}, \vec{F} = q\vec{E}$
  - work done on the charge $W = qV, W = qEd, K = \frac{1}{2}mv^2$
- model qualitatively and quantitatively the trajectories of charged particles in electric fields and compare them with the trajectories of projectiles in a gravitational field
- analyse the interaction between charged particles and uniform magnetic fields, including: (ACSPH083)
  - acceleration, perpendicular to the field, of charged particles
  - the force on the charge $F = qv_{\perp}B = qvB\sin\theta$
- compare the interaction of charged particles moving in magnetic fields to:
  - the interaction of charged particles with electric fields
  - other examples of uniform circular motion (ACSPH108)

THE MOTOR EFFECT

Inquiry question: Under what circumstances is a force produced on a current-carrying conductor in a magnetic field?

- investigate qualitatively and quantitatively the interaction between a current-carrying conductor and a uniform magnetic field $F = lI_{\perp}B = lIB\sin\theta$ to establish: (ACSPH080, ACSPH081)
  - conditions under which the maximum force is produced
  - the relationship between the directions of the force, magnetic field strength and current
  - conditions under which no force is produced on the conductor
- conduct a quantitative investigation to demonstrate the interaction between two parallel current-carrying wires
- analyse the interaction between two parallel current-carrying wires $\frac{F}{l} = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{r}$ and determine the relationship between the International System of Units (SI) definition of an ampere and Newton's Third Law of Motion (ACSPH081, ACSPH106)

ELECTROMAGNETIC INDUCTION

Inquiry question: How are electric and magnetic fields related?

- describe how magnetic flux can change, with reference to the relationship $\Phi = B_{\parallel}A = BA\cos\theta$ (ACSPH083, ACSPH107, ACSPH109)
- analyse qualitatively and quantitatively, with reference to energy transfers and transformations, examples of Faraday's Law and Lenz's Law $\varepsilon = -N\frac{\Delta\Phi}{\Delta t}$, including but not limited to: (ACSPH081, ACSPH110)
  - the generation of an electromotive force (emf) and evidence for Lenz's Law produced by the relative movement between a magnet, straight conductors, metal plates and solenoids

– the generation of an emf produced by the relative movement or changes in current in one solenoid in the vicinity of another solenoid

- analyse quantitatively the operation of ideal transformers through the application of: (ACSPH110)
  – $\frac{V_p}{V_s} = \frac{N_p}{N_s}$
  – $V_p I_p = V_s I_s$
- evaluate qualitatively the limitations of the ideal transformer model and the strategies used to improve transformer efficiency, including but not limited to:
  – incomplete flux linkage
  – resistive heat production and eddy currents
- analyse applications of step-up and step-down transformers, including but not limited to:
  – the distribution of energy using high-voltage transmission lines

APPLICATIONS OF THE MOTOR EFFECT

Inquiry question: How has knowledge about the Motor Effect been applied to technological advances?

- investigate the operation of a simple DC motor to analyse:
  – the functions of its components
  – production of a torque $\tau = nIA_{\perp}B = nIAB\sin\theta$
  – effects of back emf (ACSPH108)
- analyse the operation of simple DC and AC generators and AC induction motors (ACSPH110)
- relate Lenz's Law to the law of conservation of energy and apply the law of conservation of energy to:
  – DC motors and
  – magnetic braking

ELECTRIC FIELDS & FORCES

OUTCOMES

- investigate and quantitatively derive and analyse the interaction between charged particles and uniform electric fields, including: (ACSPH083)
  – electric field between parallel charged plates $E = \frac{V}{d}$
  – acceleration of charged particles by the electric field $\vec{F}_{\text{net}} = m\vec{a}, \vec{F} = q\vec{E}$
  – work done on the charge $W = qV, W = qEd, K = \frac{1}{2}mv^2$
- model qualitatively and quantitatively the trajectories of charged particles in electric fields and compare them with the trajectories of projectiles in a gravitational field

ELECTRIC FORCE AND ELECTRIC FIELD

FORCE BETWEEN CHARGED OBJECTS

- **Charged objects** exert a **force** on each other due to their charge.
  - This force is called the **electric force.**
  - This electric force **acts at a distance** and is mediated by an **electric field**.

Charge (q) → Electric Field (E) → Force ($F = qE$) → Work ($W = Fd = qEd$) → Kinetic ($\Delta K = W = qEd$) or Potential ($V = \frac{W}{q}$ or $W = qV$)

- A **charge** will create an **electric field** around itself.
  - Another charge will experience an **electric force from the electric field.**
  - If the other charge moves within the electric field, the **electric force will do work on it and change its energy**. (Kinetic or potential)

ELECTRIC FIELD, E (NC^{-1})

The **electric field mediates the electric force**. It is a region of influence around electric charges in which other charges experience an electric force.

The **strength** of an **electric field E** at a point in space is defined as the **electric force F** acting on a **positive test charge** placed at that point **divided** by the **magnitude of the test charge q**.

$$E = \frac{F}{q}$$

Where:

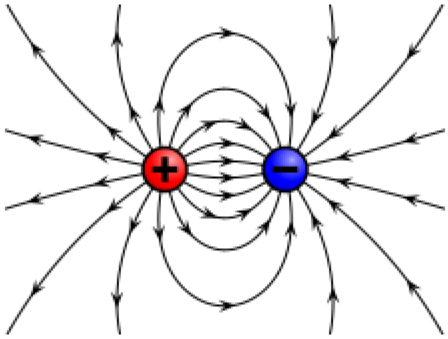
- E = external electric field strength (NC^{-1})
- F = force on positive test charge (N)
- q = amount of charge on positive test charge

If the field is known, the force on any charge in that field can be found:

$$F = qE$$

REPRESENTATION OF ELECTRIC FIELDS

- Electric fields are represented using field lines
- **Electric field lines** show two things:
  - The direction of the field, which is the **direction of force** on a **small positive test charge**.
  - The **strength of the field** indicated by the **density** of field lines.



FORCE BETWEEN TWO-POINT CHARGES: COULOMB'S LAW

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

Where:

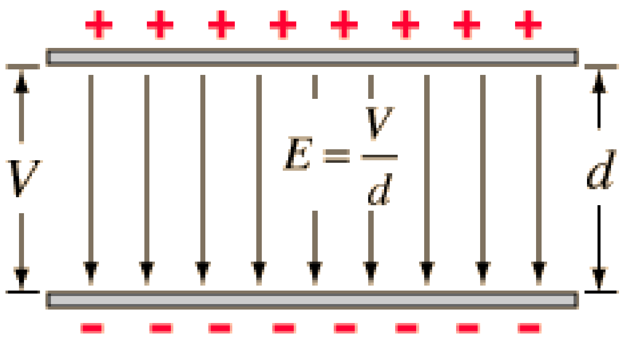
- $q_1 q_2$ are the two charges (C)
- r = distance between the two charges (m)
- $\varepsilon_0 = 8.85 \times 10^{-12}$ is the permittivity of free space. It depends on the material surrounding the charge.

The force is:

- **Attractive** for **opposite charges**
- **Repulsive** for **like charges**

THE ELECTRIC FIELD BETWEEN PARALLEL PLATES

If two conducting parallel plates are connected to a battery, the plates will become charges and an electric field will form in-between them.



The electric field formed between the plates is **uniform** and its magnitude is equal to the voltage applied to the plates divided by the distance between them.

$$E = \frac{V}{d}$$

Where:

- E = Electric field strength (N/C or V/m)
- V = Potential difference or voltage across parallel plates (V)
- d = Distance of separation of the parallel plates (m)

WORK AND ELECTRIC POTENTIAL

ELECTRICAL POTENTIAL ENERGY

Electrical potential energy (U) is defined as the potential energy of a charge due to its **position in an electric field**.

- If the electric potential energy of a charge is U, this means the electric field could do an amount of work on the charge equal to U.
- However, we have the freedom to set where U=0 and so we normally only consider charges in electrical potential energy as ΔU as the charge changes position in the field.

The work done by the electric field on a charge is equivalent to the negative of the change in electrical potential energy of the charge:

$$W = \Delta KE = -\Delta U$$

WORK DONE BY AN ELECTRIC FIELD

How the work is done by the electric field and how energy and velocity changes:

Electric Field is applied (E)

Electric force acts on the charge ($F = qE$)

The charge accelerates and increases velocity ($a = \frac{F}{m}, v = u + at$)

Kinetic energy increases, potential energy decreases

Force & Displacement	Work	Kinetic Energy	Velocity	Potential Energy
Same direction	Positive work	Increases	Increases	Decreases
Opposite direction	Negative work	Decreases	Decreases	Increases

Work being done on an object changes its energy: $W = qEd = \Delta K$

POTENTIAL DIFFERENCE (V) [V]

Potential difference or **voltage (V)** is defined as the **work done per unit charge** in moving a charge **between two points**.

$$V = \frac{W}{q}$$

Where:

- V = potential difference (V or JC^{-1})
- W = work done to move charge between two points in a field (J)
- q = magnitude of charge (C)

1 joule of work is done on every coulomb of charge that passes between two points.

$$\Delta K = W = qEd = qV$$

MOTION OF CHARGES IN ELECTRIC FIELDS

ACCELERATION OF CHARGES IN ELECTRIC FIELDS

$$\vec{F}_{net} = m\vec{a}$$

$$\therefore a = \frac{F}{m} = \frac{qE}{m}$$

The **charge-to-mass** $\left(\frac{q}{m}\right)$ ratio **describes how much charge a particle has per kilogram of mass**, in coulombs per kilogram (Ckg^{-1}).

ANALYSING MOTION IN TWO DIMENSIONS

- **Perpendicular** to the electric field: **no acceleration** ($a_\perp = 0$)
- **Parallel** to the electric field: **acceleration** ($a_\parallel = \frac{q}{m}E$)

Perpendicular Direction ($\perp$)	Parallel Direction ($\parallel$)
$\boldsymbol{v_\perp = u_\perp}$	${v_\parallel}^2 = {u_\parallel}^2 + 2a_\parallel \Delta s_\parallel$
$\boldsymbol{\Delta s_\perp = u_\perp t}$	$\Delta s_\parallel = u_\parallel t + \frac{1}{2}{u_\parallel}^2$
	$v_\parallel = u_\parallel + a_\parallel t$

MAGNETIC FIELDS & FORCES

OUTCOMES

- analyse the interaction between charged particles and uniform magnetic fields, including: (ACSPH083)
  - acceleration, perpendicular to the field, of charged particles
  - the force on the charge $F = qv_{\perp}B = qvB\sin\theta$
- compare the interaction of charged particles moving in magnetic fields to:
  - the interaction of charged particles with electric fields
  - other examples of uniform circular motion (ACSPH108)

MAGNETIC FIELD (B)

A magnetic field is a model used to explain the **region of influence** within which a magnetic force may exist.

- Magnetic fields exist around **magnets** and around **moving charges**
- The unit of magnetic field strength is the **Tesla, T.**

Moving charge (current) or permanent magnet → Magnetic Field → Magnetic force (If there are other magnets or moving charges present)

PLOTTING MAGNETIC FIELD LINES

Magnetic field lines can be plotted using magnetic compasses. The direction of magnetic field is defined as the **direction of force on a north magnetic pole**.

- The **north pole** of a compass needle will hence **always point in the same direction as the magnetic field** at a given point.
- Magnetic field lines **leave the north pole** of the magnet and **enter the south pole** of the magnet.

Magnetic field lines show two things:

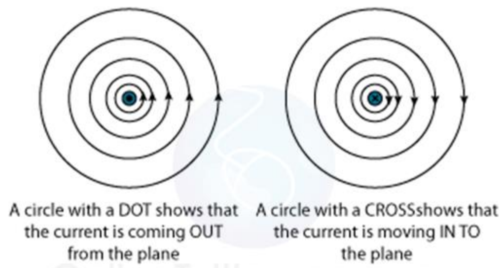
1) **The direction of a force on a north magnetic pole**, indicated by an arrow on the magnetic field lines.
2) **The strength of the magnetic field** indicated by the density of magnetic field lines.

LONG STRAIGHT CURRENT CARRYING CONDUCTOR

MAGNETIC FIELD LINES AROUND A LONG STRAIGHT CONDUCTOR

A plane view of the magnetic field of a long, straight current carrying conductor is shown below:

- **Arrow head**: Coming **out of page**
- **Arrow fletching**: Going **into the page**

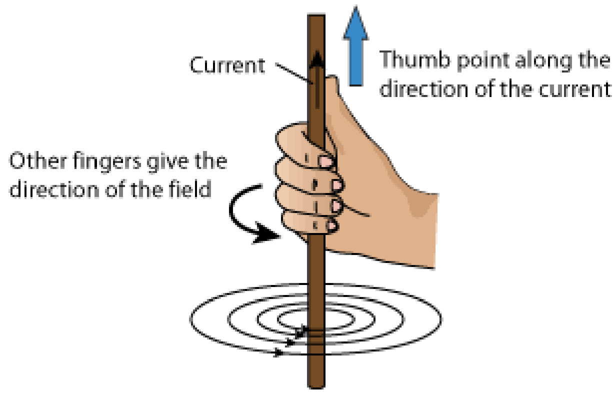


Field Pattern of Straight Wire

DETERMINING THE DIRECTION OF A MAGNETIC FIELD – RIGHT HAND GRIP RULE (RHGP)

*"Grasp the wire with the **right hand** so that the **thumb** points in the direction of the **(conventional) current**. The **curled fingers of the right-hand** point in the direction of the **magnetic field around the wire**"*

- Thumb: Direction of conventional current
- Fingers: Direction of magnetic field lines



CALCULATING THE MAGNITUDE OF A MAGNETIC FIELD

The **strength** of the **magnetic field** formed by a current carrying conductor depends on:

- The **magnitude of the current**. The strength of the magnetic field will increase as the magnitude of the current increases.
- The **distance from the wire**. The strength of the magnetic field will decrease as the distance from the wire increases.

$$B = \frac{\mu_0 I}{2\pi r}$$

Where:

- B: magnetic field strength (T)
- $\mu_0 = 4\pi \times 10^{-7}\ Tm/A$ is the magnetic **permeability of free space.** (Depends on materials around it)
- I: Current through the wire (A)
- r: Distance from the wire (m)

MAGNETIC FIELDS AROUND TWO LONG PARALLEL CONDUCTORS

- Currents flowing in the **opposite direction repel**.
- Currents flowing in the **same direction attract**.

When the magnetic fields from two different current carrying conductors interact with each other, the **resultant magnetic field will change in magnitude and direction**. The total field will be a **vector sum** of the two fields.

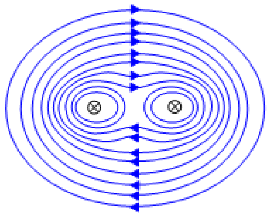


Figure 2(a)

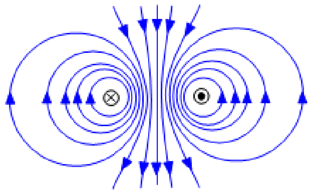
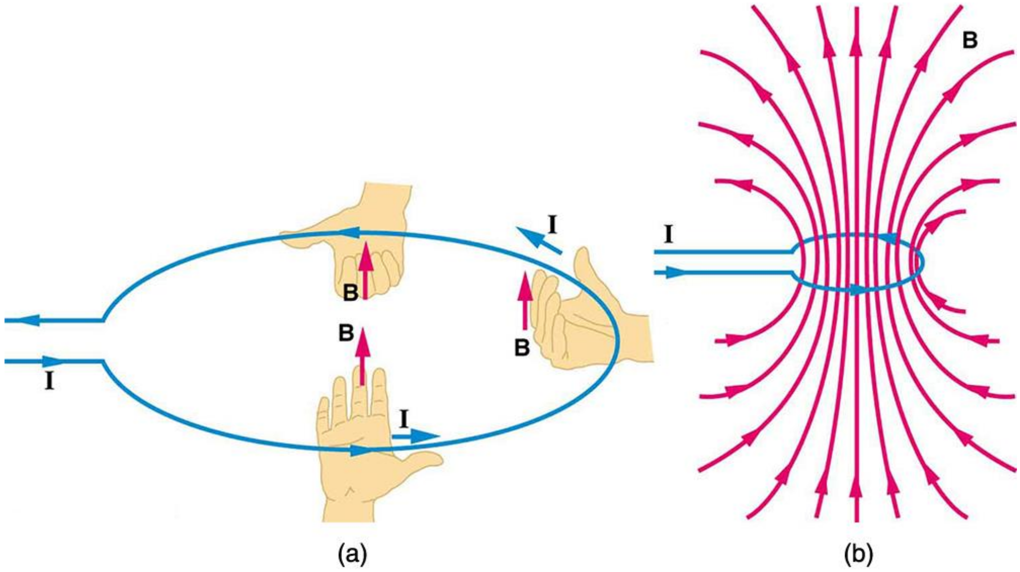


Figure 2(b)

- If the two magnetic fields are in the **same direction**, then the resultant magnetic field will increase in magnitude: $B_{Total} = B_1 + B_2$
- If the two magnetic fields are in the **opposite direction**, then the resultant magnetic field will decrease in magnitude: $B_{Total} = B_1 - B_2$

COILS OF WIRE

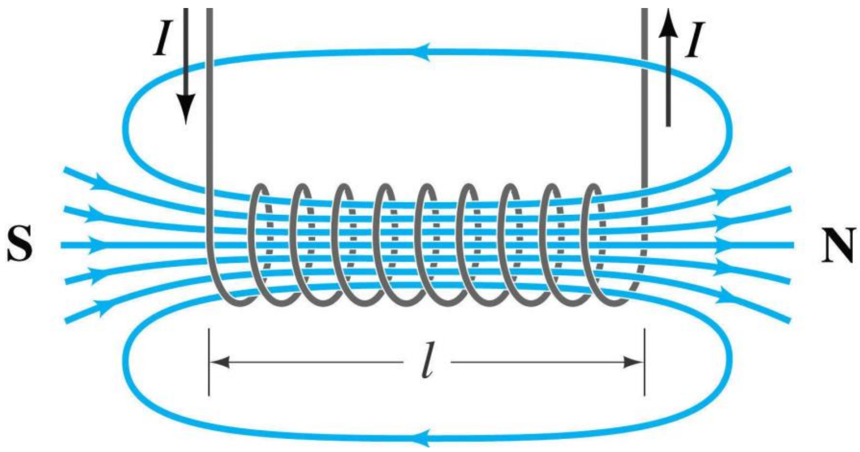
MAGNETIC FIELDS AROUND A CIRCULAR LOOP OF WIRE



- The field around the wire can be determined by using the **Right-Hand Grip Rule**.
- The field concentrates inside the loop to produce a stronger field with a **north pole** and a **south pole** as pictured. Can be determined through the **Right-Hand Coil Rule.**

MAGNETIC FIELD OF A SOLENOID

- A solenoid is a coiling of wire in the shape of a cylinder. It is composed of a number of 'turns' (n or N).
- The **magnetic field of a solenoid** resembles that of the long bar magnet, and behaves as if it has a north pole at one end and a south pole at the other.
- If the turns are close together and the solenoid is long relative to its diameter, then the **magnetic field within it is uniform and parallel** to its **axis except near the ends**.



$$B = \frac{\mu_0 NI}{L}$$

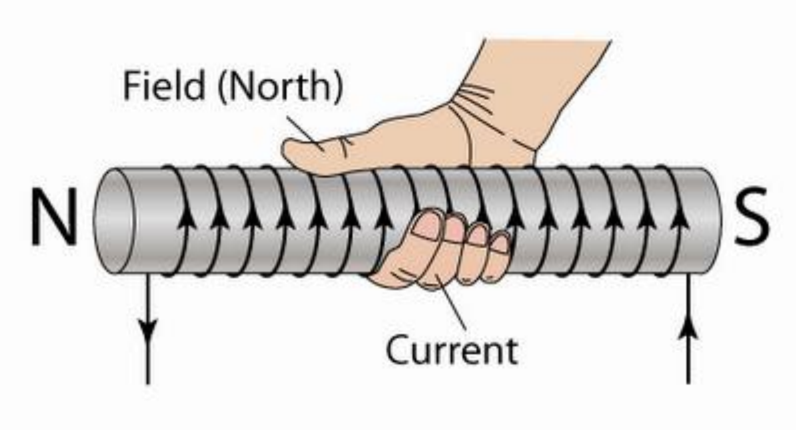
Where:

- B: magnetic field strength (T)
- $\mu_0 = 4\pi \times 10^{-7}\ Tm/A$ is the magnetic **permeability of free space.** (Depends on materials around it)
- I: Current through the wire (A)
- N: number of turns of the solenoid
- L: length of the solenoid

RIGHT HAND COIL RULE (RHCR)

The direction of the magnetic field in a solenoid or coil is given by the **Right-Hand Coil Rule**.

*"Grasp the loop so that the **curled fingers of the right-hand** point in the **direction of the current**. The **thumb of the right-hand points in the direction of the magnetic fields** within the loop of wire."*



MAGNETIC FORCE ON A MOVING CHARGE

The magnitude of the magnetic force is given by:

$$F = qv_{\perp}B = qvB\sin\theta$$

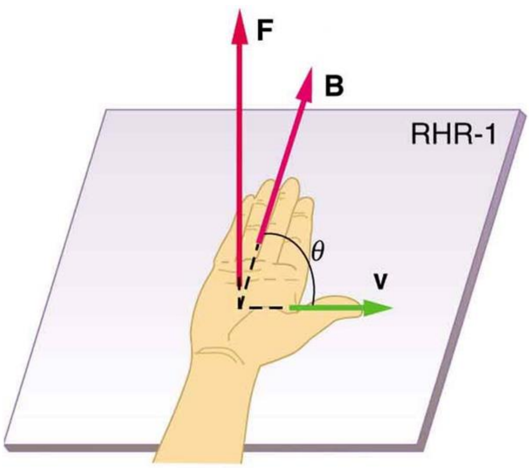
Where:

- q: is the magnitude of the charge (C)
- B: is the magnitude of the external magnetic field (T)
- $v_{\perp}$: is the component of the velocity of the particle **perpendicular to the magnetic field.** (ms^{-1})
- θ: Angle between velocity vector and magnetic field vector

RIGHT-HAND PALM RULE (RHPR)

The **direction** of the magnetic force is determined by using the **Right-Hand Palm Rule**.

- Fingers: Direction of magnetic field
- Thumb: Direction of velocity
- Palm: Direction of force



$F = qvB\sin\theta$

F $\perp$ plane of **v** and **B**

PATH OF A CHARGED PARTICLE

The path of a charge particle moving perpendicular to its uniform magnetic field is **circular**.

- A **positive charge** q, moving with a velocity v at **right angles** to a uniform magnetic field experiences a force of magnitude ($F = qvB$)
- The direction of the force is determined using the RHPR. The direction of the force F is **always perpendicular** to the particle's **velocity.**
- In a uniform magnetic field, this causes the path of the particle to curve at a constant rate. The particle follows **uniform circular motion**.

Magnetic force = Centripetal Force

$$qvB = \frac{mv^2}{r}$$

$$\therefore qB = \frac{mv}{r}$$

Hence, the radius of the path of the particle is

$$r = \frac{mv}{qB}$$

THE MOTOR EFFECT

OUTCOMES

Inquiry question: Under what circumstances is a force produced on a current-carrying conductor in a magnetic field?

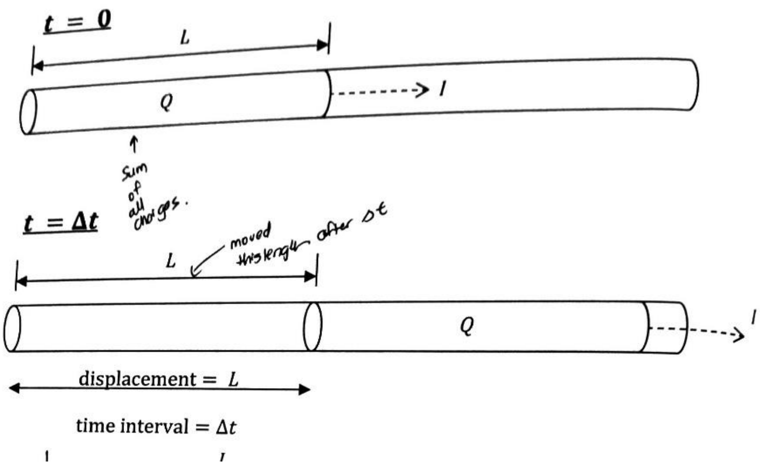
- investigate qualitatively and quantitatively the interaction between a current-carrying conductor and a uniform magnetic field $F = lI_{\perp}B = lIB\sin\theta$ to establish: (ACSPH080, ACSPH081)
  - conditions under which the maximum force is produced
  - the relationship between the directions of the force, magnetic field strength and current
  - conditions under which no force is produced on the conductor
- conduct a quantitative investigation to demonstrate the interaction between two parallel current-carrying wires
- analyse the interaction between two parallel current-carrying wires $\frac{F}{l} = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{r}$ and determine the relationship between the International System of Units (SI) definition of an ampere and Newton's Third Law of Motion (ACSPH081, ACSPH106)

THE MOTOR EFFECT

MAGNETIC FORCE ON A CURRENT CARRYING CONDUCTOR

A **current** will experience a **force** in a **magnetic field**. This will be the **sum of the force on all moving charges** that make up the current.

If you place a **current carrying conductor in a magnetic field**, the conductor **experiences a magnetic force** due to the **interaction between the existing magnetic field** and **magnetic field generated by the current**. This is known as the **motor effect**.



- The diagram shows a block of charge Q. In the time Δt the block of charge travels distance L. Therefore its speed is $v = \frac{L}{\Delta t}$.
- Current is defined as the total charge passing a point per time, $I = \frac{q}{\Delta t}$

Force on a charge Q moving with the velocity of v in the magnetic field B has the magnitude $F = Qv_{\perp}B = QvB\sin\theta$

Therefore, the force F on a conductor of length L carrying current I at an angle of θ to the magnetic field B is:

$$F = BI_{\perp}L = BIL\sin\theta$$

Where:

- F: force on current carrying conductor in a magnetic field
- B: the strength of the magnetic field
- L: the length of the conductor **inside** the magnetic field
- θ: Angle between the current vector and magnetic field vector

Since the current consists of the flow of positive charges, the **direction** of the force on a current carrying conductor can be found by using the **RHPR**.

FORCE BETWEEN PARALLEL CURRENT CARRYING CONDUCTORS

AMPERE'S FORCE LAW

A **current** produces a **magnetic field**, and hence a current-carrying conductor experiences a **force** when its field interacts with existing magnetic fields. Therefore, parallel current carrying conductors must **exert forces** on **each other**.

Andre Ampere measured the forces between the current carrying wires by an **experimental arrangement** known as the **current balance.** Ampere was able to determine the variation in force between the current carrying wires as different quantities were changed.

Ampere found that the magnitude of the force was:

- **Directly proportional** to the products of the currents
- **Directly proportional** to the common parallel length of the wires
- **Inversely proportional** to their separation distance

$$\frac{F}{L} = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{d}$$

Where:

- F: Force on wires (N)
- L: Length of wires (m)
- $\mu_0 = 4\pi \times 10^{-7}\ Tm/A$ is the magnetic **permeability of free space.** (Depends on materials around it)
- $I_1 I_2$: current of wire 1 and wire 2 (A)
- d: Distance between wires (m)

Newton's third law: $\overrightarrow{F_{12}} = -\overrightarrow{F_{21}}$

- The force between conductors with currents travelling in the **same direction** is **attractive**.
- The force between conductors with currents travelling in the **opposite direction** is **repulsive**.

DEFINITION OF THE AMPERE

Ampere's parallel wire experiment is how the value of the **Ampere** was defined in 1954.

"The ampere (1A) is that constant current which, if maintained in two straight parallel conductors of infinite length, of negligible circular cross-section, and placed 1 metre apart in a vacuum, would produce between these conductors a force equal to 2×10^{-7} *newton per metre of length."*

SECOND-HAND INVESTIGATION: FORCE BETWEEN PARALLEL CONDUCTORS

1) Plot the data on an **appropriate** graph and draw a line of best fit
2) Calculate the gradient
3) Express the gradient in terms of constants & control variables.
4) Use information in Step 2 and 3 to find the unknown

F VS I

Plotting F vs I would result in a parabolic curve

$$F = \frac{\mu_0 L}{2\pi r} \times (I^2)$$

F VS I^2

Straight line where the gradient represents: $m = \frac{\mu_0 L}{2\pi r}$

F VS D

Plotting F vs d would result in a hyperbola curve.

$$F = \frac{\mu_0 L I^2}{2\pi} \times \frac{1}{d}$$

F VS D^{-1}

Plotting F vs $\frac{1}{d}$ would result in a straight line where the gradient represents: $m = \frac{\mu_0 L I^2}{2\pi}$

APPLICATIONS OF THE MOTOR EFFECT

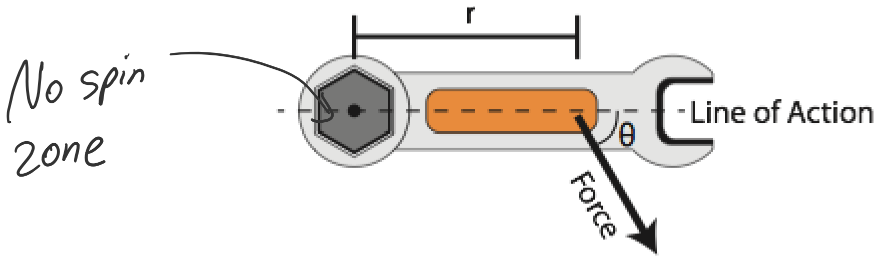
OUTCOMES

Inquiry question: How has knowledge about the Motor Effect been applied to technological advances?

- investigate the operation of a simple DC motor to analyse:
  - the functions of its components
  - production of a torque $\tau = nIA_{\perp}B = nIAB\sin\theta$

TORQUE

Torque is defined as the **tendency of a force** to **rotate an object about an axis**.



$$\boldsymbol{\tau = r_{\perp}F = rFsin\theta}$$

Where:

- τ: Torque
- $r_{\perp}$: Perpendicular distance from the axis of rotation of F

CONDITION FOR MAXIMUM TORQUE

For a given force, the torque produced by the force is **maximum** when the force is applied **perpendicular** to the line joining the force and the pivot point. (When $\theta = 90°$)

TORQUE ON A CURRENT LOOP

A **force** is exerted on a **current carrying conductor** when the conductor is **placed in an external magnetic field**. **Torque** is exerted on a **current loop** placed in a **magnetic field**.

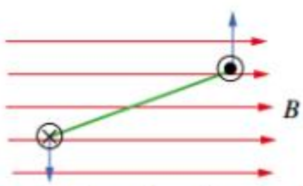


FIGURE 8.1.5 Top view of a coil turning in a magnetic field. The force experienced by each end of the coil is always perpendicular to the field.

Since a force is acting at a distance from an **axle**, a torque is created as shown in the diagram above.

ANALYSIS OF A CURRENT LOOP

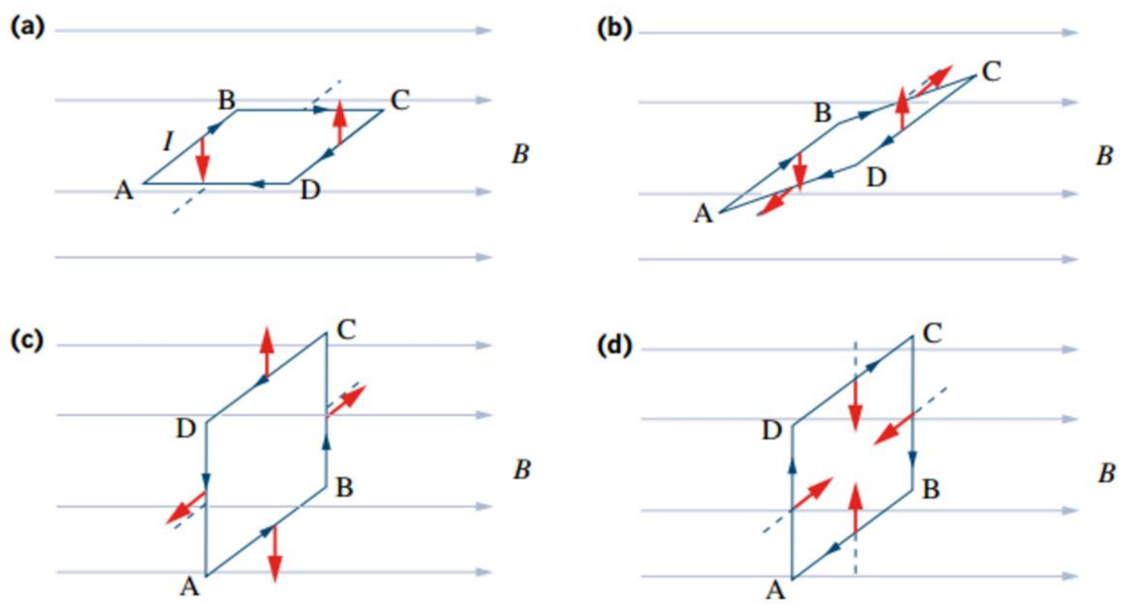
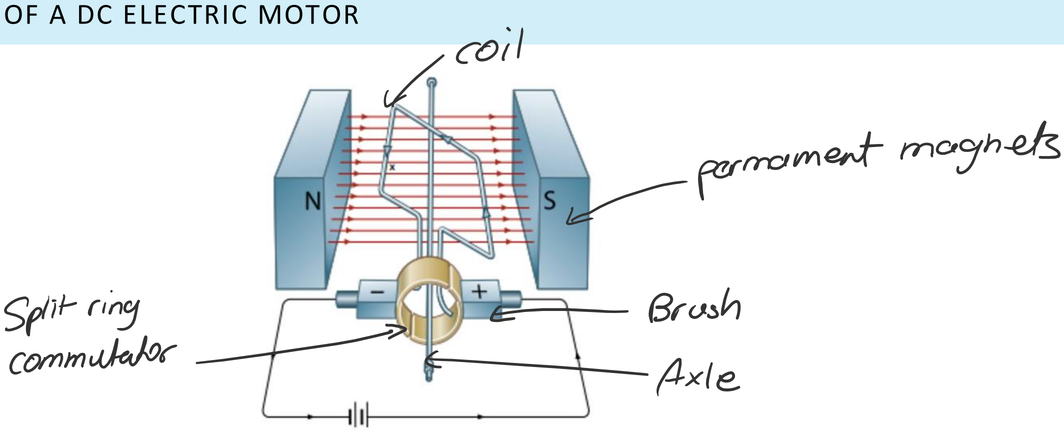


FIGURE 8.1.2 The magnetic force acting on each side of a current-carrying square wire coil in a magnetic field.

- It is fastest when the coil is perpendicular to the magnetic field. (C)
- Torque is maximised when the coil is horizontal to the magnetic field (A)
- Torque is minimised (0) when the coil is perpendicular to the magnetic field (C) or (D)

DC ELECTRIC MOTOR

MAIN COMPONENTS OF A DC ELECTRIC MOTOR



The main components of a DC electric motor are:

- A pair of permanent magnets or pair of electromagnets
- Armature
- Rotor coils or armature coils
- Split ring commutator
- Brushes
- Axle or shaft

THE MOTOR EFFECT IN DC MOTORS

- DC supplied to the motor
- Commutator reverse direction of current every half turn to provide AC in the coil
- AC passes through a rotor coil in an external magnetic field
- The rotor coil experiences a magnetic field ($F = LIB$)
- A torque is produced ($\tau = rFsin\theta$)
- The rotor coils rotate

ROLE OF MAIN COMPONENTS

Part		Description	Role of Part
Stator	Pair of permanent magnets	Two permanent magnets on opposite sides of the motor, with opposite poles facing each other. The poles faces are curved to fit around the armature	The magnets supply the magnetic field which interacts with the current in the rotor coils to produce the motor effect. The shape of the pole faces makes the magnetic field almost uniformly radial where the coil passes
	Pairs of electromagnetic coils	Each stator coil is wound on a soft iron core attached to the causing of the motor. The coils are shaped to fit around the armature	Each opposed pair of stator coils produces a magnetic field similar to that provided by a pair of permanent magnets. The iron core concentrates the field.
Rotor	Armature	The armature consists of a **cylinder of laminated iron** mounted on an axle. Often there are longitudinal grooves into the coils are wound	The armature carries the rotor coils. The iron core greatly concentrates the external magnetic field, increasing the torque on the armature. The laminations reduce eddy currents which might otherwise overheat the armature.
	Rotor Coils	There may only be one, in a very simple motor, or several coils, usually of several turns of insulated wire, wound onto an armature. The ends of the coils are connected to bars on the commutator.	The coils provide torque, as the current passing through the coils interact with the magnetic field. As the coils are mounted firmly on the rotor, any torque acting on the coils is transferred to the rotor and then to the axle. Each coil is positioned at right angles to the field, to maximise torque.
	Axle	A cylindrical bar of hardened steel passing through the centre of the armature and the commutator.	The axle provides a centre of rotation for moving parts of the motor. Useful work can be extracted from the motor via a pulley or a cog mounted on the axle.
Contacts	Split-ring commutator	The commutator is a broad ring of metal mounted on the axle at one end of the armature and cut into an even number of separate bars (two in a simple motor). Each opposite pair of bars is connected to one coil	The commutator provides points of contract between the rotor coils and the external electric circuit. It serves to reverse the direction of current flow in each coil every half-revolution. This ensures that the torque on each coil is always in the same direction.
	Brushes	Compressed carbon blocks, connected to the external circuit, mounted on opposite sides of the commutator and spring-loaded to make close contact with the commutator bars	The brushes are fixed position electrical contacts between the external circuit and the rotor coils. Their position brings them into contract with both ends of each coil simultaneously.

MAINTAINING TORQUE IN THE SAME DIRECTION

In order for a **DC motor** to rotate in one direction continuously, torque produced must be maintained in the same direction. Maintaining torque direction is achieved by **reversing the direction of the current** once every half turn whilst the direction of the magnetic field in the stator stays constant. This is done through a **split-ring commutator**.

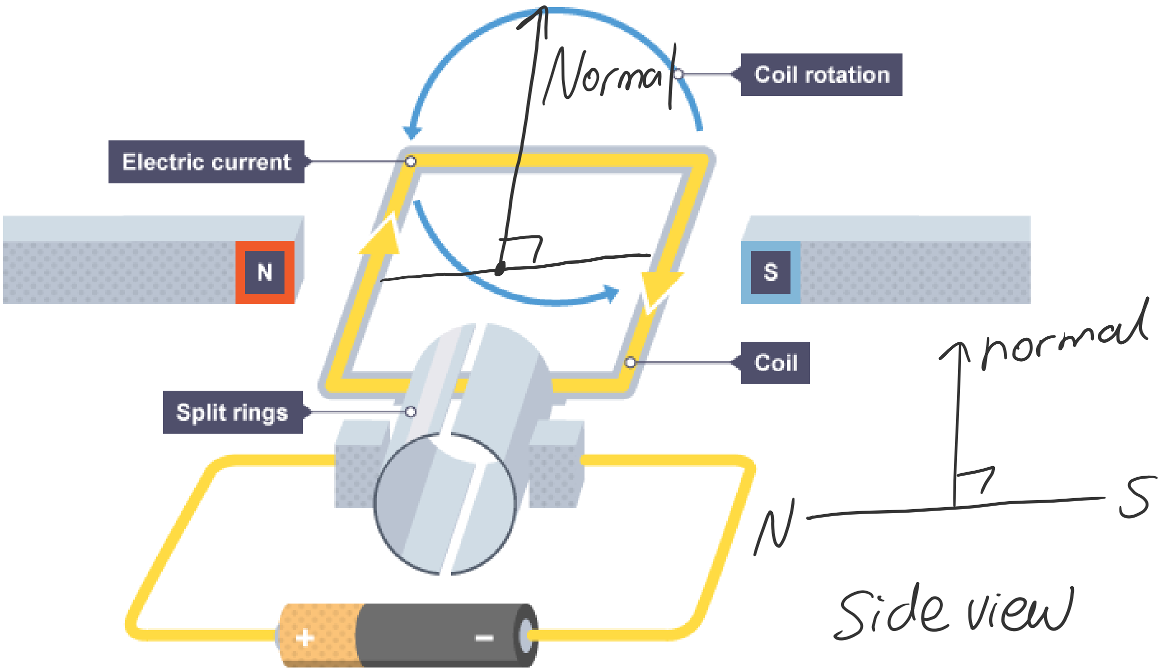
The torque **fluctuates between 0 and a maximum value** and is always in the same direction. It is a positive sine curve.

PROBLEMS WITH A SIMPLE BRUSHED DC MOTOR

The main problems with a brushed DC motor are:

- The **brushes wear out** and need replacement.
- The **commutator gets worn**, which shortens the life of the motor.
- **Sparking at the commutator** can also be dangerous if there is flammable gases in the air

TORQUE IN A RECTANGULAR LOOP OF WIRE



The torque acting on n turns of the coil is given by:

$$\tau = nIAB_{\perp}B = nIABsin\theta$$

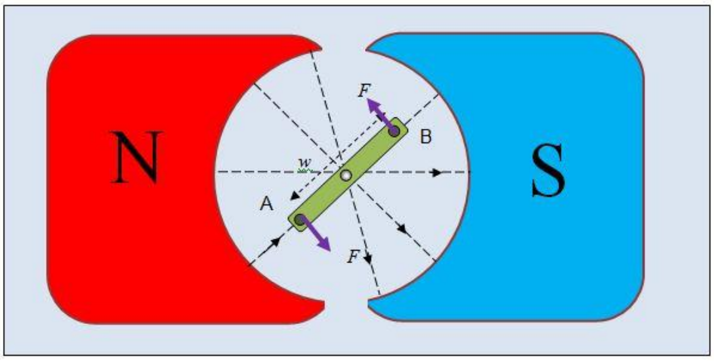
Where:

- n: number of loops in the coil
- B: the magnetic field strength
- I: current through the coil
- A: area of the coil
- θ: the angle between the **normal of the coil area** and the direction of the magnetic field.

Torque is **maximum** when the **plane of the coil is parallel to the magnetic field.**

METHODS OF IMPROVING TORQUE CHARACTERISTICS

RADIAL MAGNETS

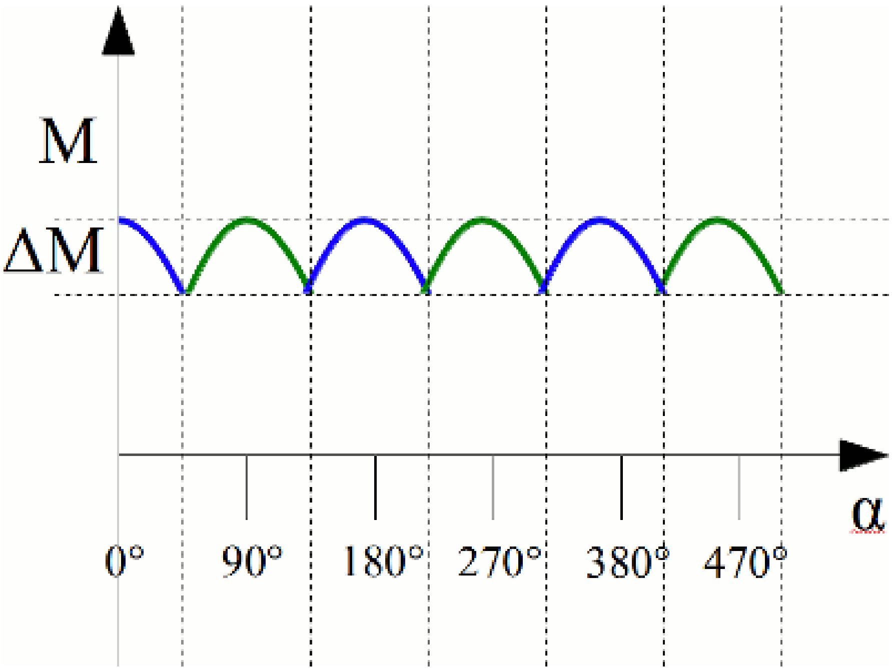


The curved permanent magnets produce a **radial magnetic field**. This ensures that the plane of the coil is always perpendicular to the magnetic field, producing the maximum torque at any angle.

A SET OF PERPENDICULAR COILS

A steadier torque can be obtained by using a **perpendicular pair of rotor coils**.

- When one of the coils produces maximum torque, the other produces 0 torque
- The resultant torque will be steadier and fluctuate less



Coil 1 – Blue

Coil 2 - Green

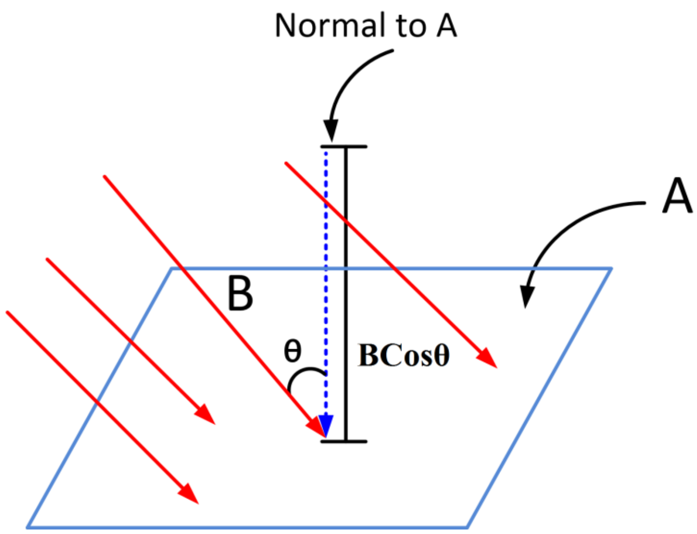
FARADAY’S AND LENZ’S LAWS

OUTCOMES

- describe how magnetic flux can change, with reference to the relationship $\Phi = B_{\parallel}A = BA\cos\theta$ (ACSPH083, ACSPH107, ACSPH109)
- analyse qualitatively and quantitatively, with reference to energy transfers and transformations, examples of Faraday’s Law and Lenz’s Law $\varepsilon = -N\frac{\Delta\Phi}{\Delta t}$, including but not limited to: (ACSPH081, ACSPH110)
  - the generation of an electromotive force (emf) and evidence for Lenz’s Law produced by the relative movement between a magnet, straight conductors, metal plates and solenoids
  - the generation of an emf produced by the relative movement or changes in current in one solenoid in the vicinity of another solenoid

MAGNETIC FLUX

Magnetic flux is visualised as a measure of the **total number of magnetic field lines that pass through a chosen surface area**.



$$\phi = B_{\parallel}A = BAcos\theta$$

Where:

- ϕ: magnetic flux in units of Weber (Wb)
- B: magnetic field strength in units of Tesla (T) for the component of **magnetic field parallel to the area’s normal vector**
- A: surface area (m^2)

As flux measures the field lines that **pass through the area**, these field lines must be perpendicular to the area itself, hence **parallel to the normal vector**.

The magnetic field strength **B** can be defined in terms of the **magnetic flux density** which is the amount of magnetic flux per unit area:

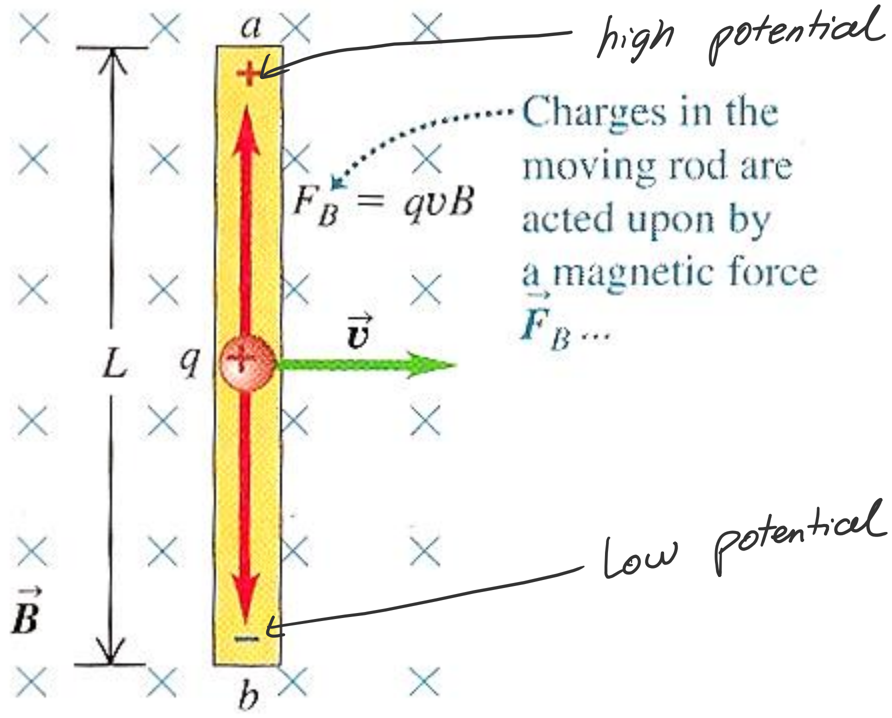
$$B = \frac{\phi}{A}$$

ELECTROMAGNETIC INDUCTION

Electromagnetic induction refers to the production of an induced EMF (electro-motive force) and (sometimes) an induced current by the use of **magnetic means only**.

- EMF can be thought as a voltage or potential difference, which is induced in a conductor as a result of the relative motion of a magnetic and a coil

CASE STUDY 1

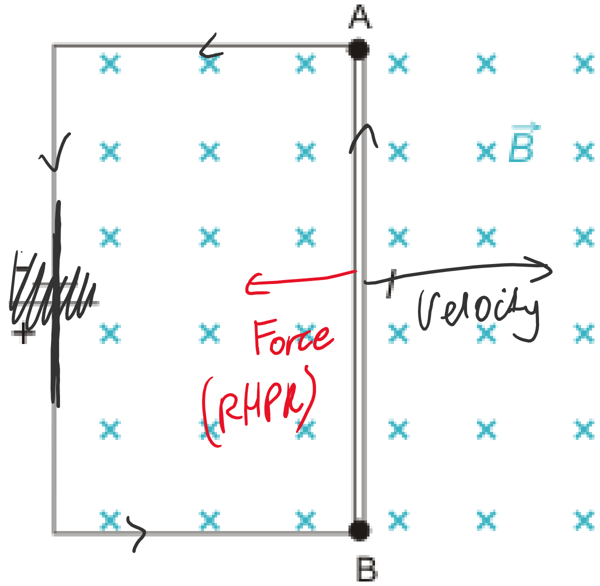


Assume that the conductor is moving perpendicular to the field:

- The electrons experience a force **downwards** (by left-hand palm rule) ($F = qvBsin\theta$)
- The positive charges experience a force **upwards** (by right-hand palm rule) but protons aren't free to move in solids.
- This creates a higher potential (more positive) at the top of the conductor.
  - The separation of charges leads to a **potential difference** (voltage) is established between the ends of the rod.
- If connected to an external circuit, the induced current (conventional current) would be anticlockwise.

CASE STUDY 2:

If there is a **closed conducting circuit, current will flow** from the higher potential end of the rod to the lower potential end through the external circuit.



http://physicstasks.eu/

There will thus be a **force due to the motor effect** on the rod ($F = BILsin\theta$). The direction of this force relative to the motion of the rod is the **opposite direction** of velocity.

When there is **movement inducing current**, there is a **force opposing motion**. This ensures that the conversation of energy in a system is true.

MICHAEL FARADAY

Michael Faraday is considered the father of the **electric generator** and the developer of the theory that resulted in the invention of the electric motor. He performed the following experiments:

1) **Induction with a magnet**. Relative motion between and magnet and a coil produces changing magnetic fields which in turn produces an electromotive force (EMF)
2) **Induction with coils**. A non-steady current in a primary coil produces changing magnetic fields which in turn induces a non-steady current in a secondary coil.

CONCLUSION OF FARADAY'S EXPERIMENTS

An **EMF** is induced in a circuit when the **magnetic flux** through the circuit **changes** with time.

- Change in magnetic flux → EMF Induced → Current Induced (If circuit is closed)

FARADAY'S LAW OF INDUCTION

An EMF is induced whenever a circuit experiences a change of magnetic flux and the **EMF induced** in a circuit is **directly proportional** to the **rate of change** of **magnetic flux** through the circuit.

$$\epsilon = -\frac{\Delta\phi}{\Delta t}$$

Where:

- $\Delta\phi$: is the change in flux through the circuit (e.g. coil)
- Δt: is the time over which the change occurred

The meaning of the **negative sign** in the equation is a consequence of **Lenz's law**. If the circuit coil contains N loops, all the same area and with the same flux threading through all the loops, the induced EMF is given by:

$$\epsilon = -\frac{N\Delta\phi}{\Delta t} = -\frac{N\Delta BAcos\theta}{\Delta t}$$

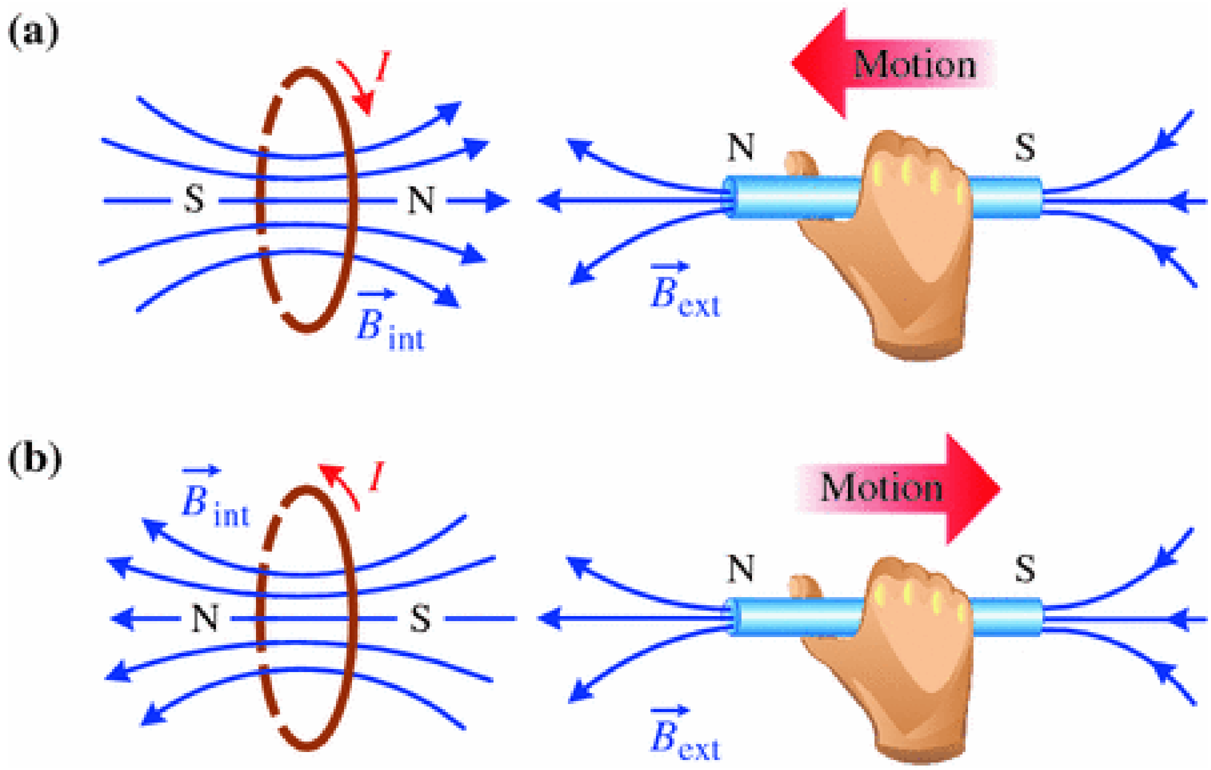
EMF can be induced in the circuit several ways:

- The magnitude of the magnetic field varies with time (magnetic moving closer or further away)
- The **area of the circuit** can change with time (coil is squashed)
- The **angle** between magnetic field and the normal to the plane changes with time (coil rotating)

LENZ'S LAW

Lenz's Law states: The direction of the induced EMF in a circuit is such that it attempts to produce a current that will create its own magnetic flux to **opposite the original change** in magnetic flux through the loop.

The **induced current tries to prevent** the **magnetic flux through the circuit changing**.



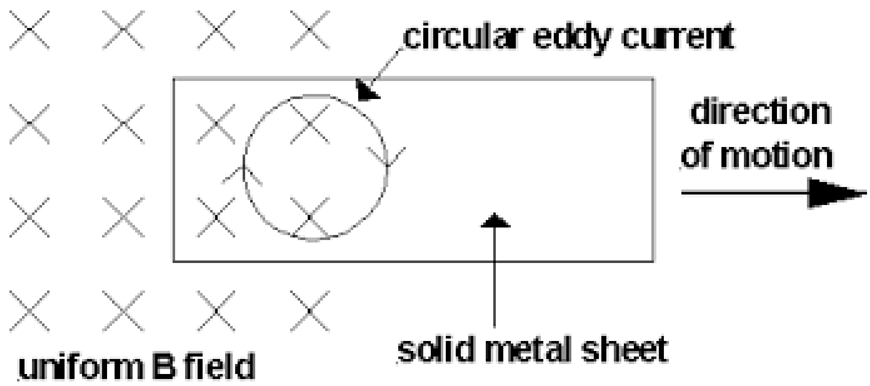
APPLICATIONS OF LENZ'S LAW

OUTCOMES

- investigate the operation of a simple DC motor to analyse:
  - effects of back emf (ACSPH108)
- analyse the operation of simple DC and AC generators and AC induction motors (ACSPH110)
- relate Lenz's Law to the law of conservation of energy and apply the law of conservation of energy to:
  - DC motors and
  - magnetic braking

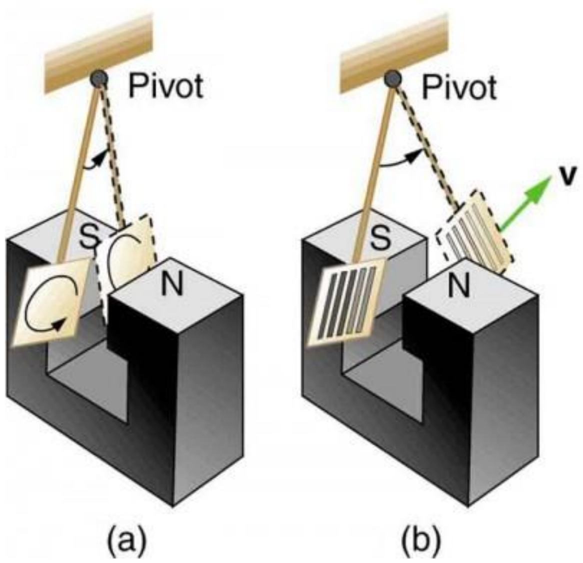
EDDY CURRENTS

Eddy currents are **circulating induced currents** in bulk pieces of metal **experiencing a change in magnetic flux**.



The bulk piece of metal forms a **closed circuit**, and allows current to flow when there is an induced EMF.

CASE STUDY: PENDULUM SWINGING IN A MAGNETIC FIELD



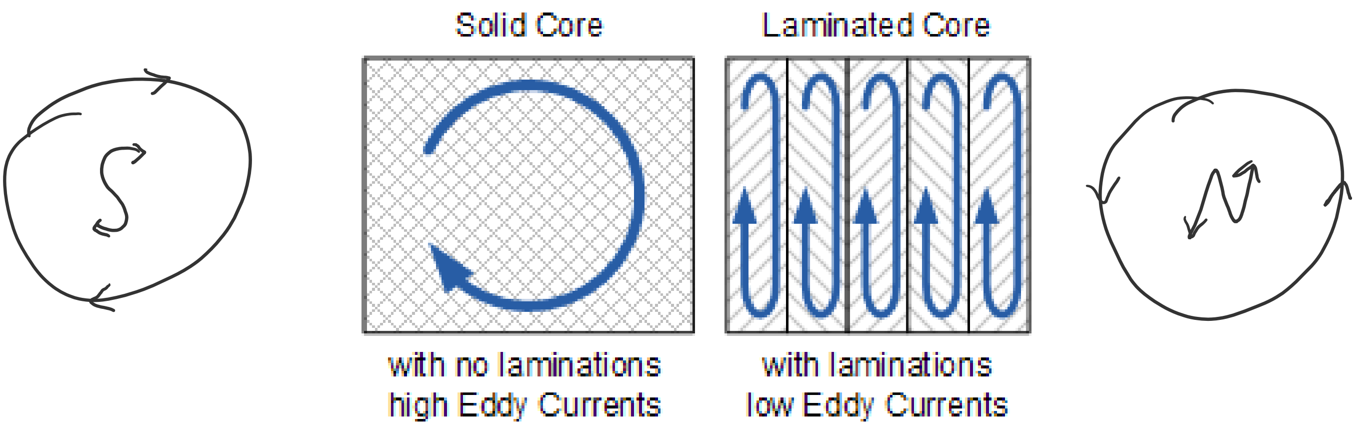
A: As the plate enters the field, the changing magnetic flux creates an induced EMF. Eddy currents (circulating free electrons in metal caused by the induced EMF) are created. According to Lenz's law, the direction of the eddy currents must oppose the change that causes them.

- The eddy currents produce **effective magnetic poles** (like a solenoid) on the plate, which are repelled or attracted by the poles of the magnet, thus giving rise to a **retarding force** that opposes the motion of the pendulum.

B: The slots cut out from the metal plate restrict the flow of eddy current with insulating air. The magnetic poles the eddy current creates are thereby weakened as are the retarding forces exerted on them by the external field. The eddy currents formed are more localised and therefore restricted.

PROBLEMS WITH EDDY CURRENTS

Eddy currents are often undesirable as they dissipate energy in the form of heat. To reduce energy loss, the moving conductor parts are often **laminated**. A conductor is laminated by slicing it into thin sections and separating each one with insulating (plastic) layers.



It is effective when the laminations are perpendicular to the plane of the coil.

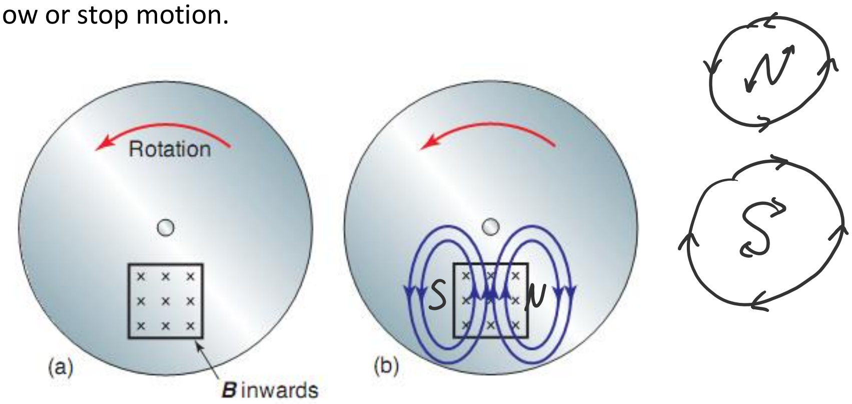
ELECTROMAGNETIC BRAKING

There are two main types of magnetic breaking:

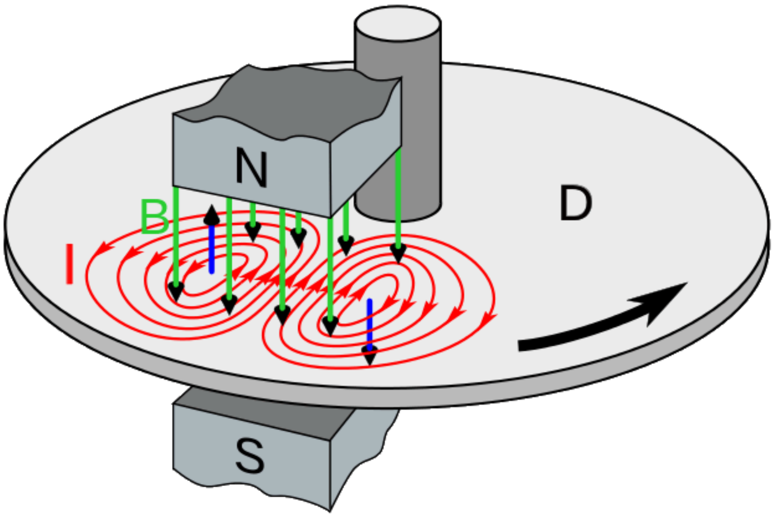
1) Electromagnetic (eddy current) braking used on metal wheels like in trains
2) Regenerative braking used in situations involving electric motors like electric cars

ELECTROMAGNETIC BRAKING

Electromagnetic braking refers to the a braking system that uses electromagnetic forces set up by eddy currents in a rotating metal plate or disc to slow or stop motion.



As the plate rotates, region 1 and region 2 of the plate experience a change in magnetic flux. Region 1 experiences an **increasing magnetic flux** and region 2 experiences a **decreasing magnetic flux.** According to Faradays and Lenz's law, an EMF will be induced such that it creates its own magnetic flux that opposes the original motion creating it.



The regions of the wheel entering and leaving the magnetic field experiences changes in magnetic flux which induces EMFs (Faraday's law: $\epsilon = -\frac{\Delta\phi}{\Delta t}$) and drive eddy currents in the metal. These eddies opposite the change in magnetic flux that induced them (Lenz's law) – they create their own magnetic poles on which the magnet exerts retarding forces.

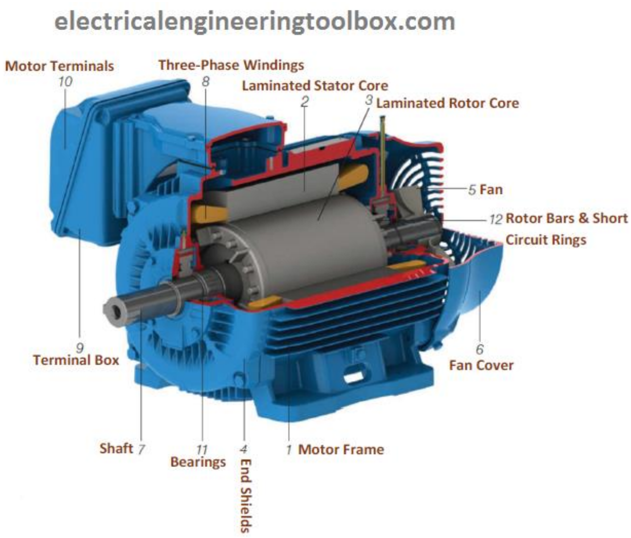
AC INDUCTION MOTORS

SINGLE PHASE VS THREE PHASE AC POWER SUPPLY

Power stations generate **three separate electrical circuits** of AC which are out of phase with each other by 120 degrees.

- Most customers receive only a single circuit of AC, called **single phase**
- However, some industrial customers receive all three AC circuits, receiving **three-phase power**
- The **induction motor** is a motor that operates on a **three-phase AC power supply**.

MAIN FEATURES OF THREE PHASE INDUCTION MOTOR



An induction motor consists of:

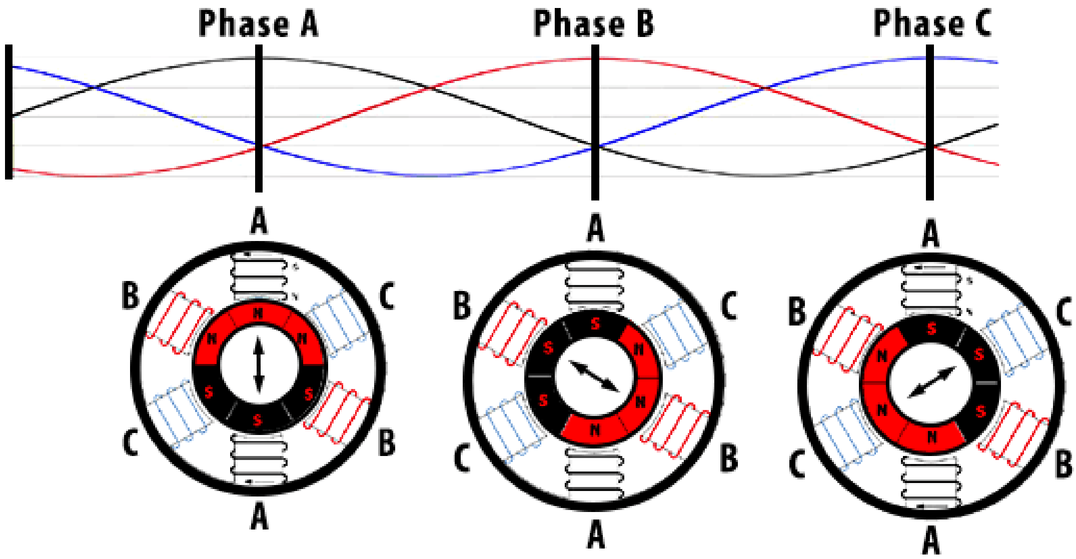
1) A stator consisting of multiple coils
2) A rotor consisting of cylinder made metal
3) An axle
4) No slip rings or brushes

PRINCIPLE OF THE AC INDUCTION MOTOR

AC induction motors work quite differently to DC motors:

- The **stator** consist of **multiple electromagnetic coils,** whereas the **rotor** is a **cylinder of metal** (called a squirrel cage rotor)
- The **stator coils** are connected to three-phase AC which produces a **rotating magnetic field**
- This induces **eddy currents** in the rotor, causing it to move due to Lenz's law.

3 PHASE INDUCTION MOTOR

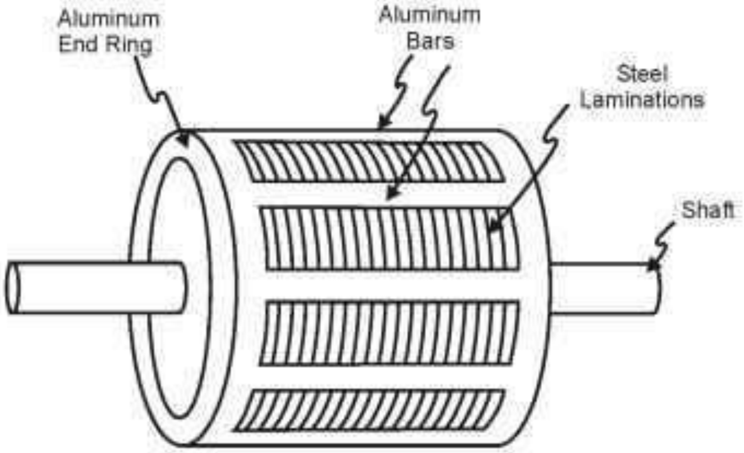


The three pairs of stator coils connected to three-phase AC power supply produces an **apparent rotating magnetic field in a clockwise direction**. The three phases have different currents at the same time, hence coils they are attached to produce a magnetic field at different times. The combined effect of the magnetic fields of these coils over time produces a rotating magnetic field.

SQUIRREL CAGE ROTOR

The rotor consists of cylindrical laminated core with parallel slots (armatures) for carrying thick bars of **copper** or **aluminium** in a physical arrangement known as a **squirrel cage**.

- The bars are connected at the ends by a ring or disk of copper or aluminium which allows **current to flow in a loop** between opposite bars.
- The **iron core** intensifies the magnetic field of the stator, inducing a larger current in the rotor, resulting in larger torque.



SUMMARY OF AC INDUCTION MOTOR

OPERATION OF THE STATOR

1) Opposing stator coils are connected to one phase of the three-phase current.
2) The AC continually changes the strength of each pair of electromagnetics
3) This **creates an apparent rotating magnetic field in the stator**.

OPERATION OF THE ROTOR

4) The rotating magnetic field **induces currents** with **opposing magnetic polarities in the bars of the squirrel cage rotor**
5) The eddy currents produce magnetic fields which interact with the rotating field of the stator to exert a torque on the rotor in the direction of rotation of the stator field
6) The rotor rotates in the same direction as the magnetic field.

An induction motor has a **fixed maximum speed**. The magnetic field of the stator rotates at the frequency of the AC power supply. Induction motors spin about 3000 revolutions a minute without a load in Australia (50 Hz x 60 seconds).

BACK EMF IN A DC MOTOR

In an electric motor, when a current is supplied to the coil placed in the magnetic field, the coil experiences torque and thus turns.

When the electric motor is turning, the coil of the armature is turning with respect to a stationary magnetic produced by the stator. The magnetic flux through the area of the armature is always **constantly changing**.

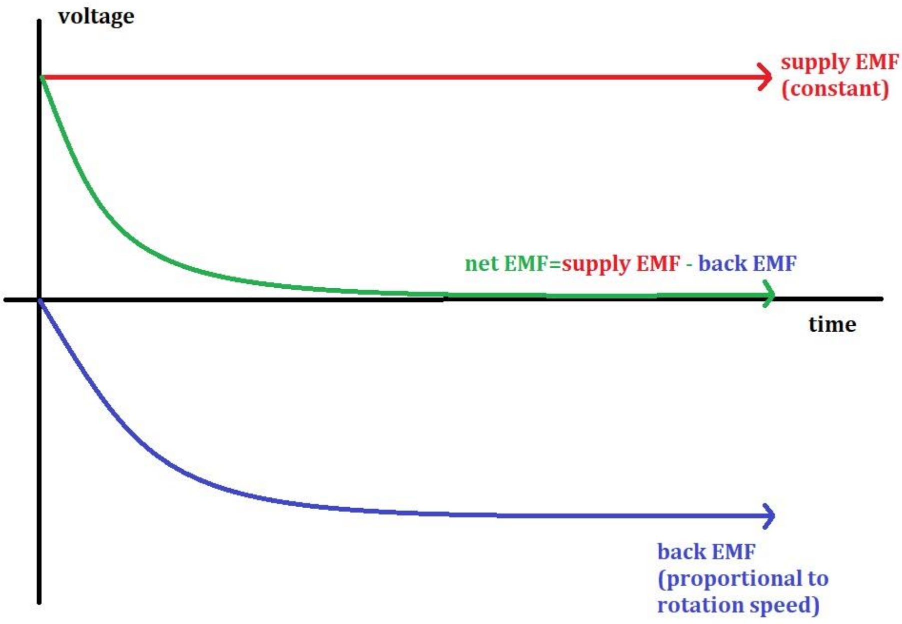
As a result, the induced EMF produced in the electric motor as a result of the change in flux is called **back EMF** as it **opposes** the supply EMF (voltage supplied by battery).

RELATIONSHIP BETWEEN BACK EMF AND CURRENT

There is **no induced current in the coil** as a result of the back EMF. Instead, the net EMF and hence the **current in the coil** is a result of the **difference** between the supply EMF and back EMF.

$$EMF_{coil} = EMF_{supply} - EMF_{back} = I_{coil}R$$

$$I = \frac{V}{R} = \frac{V_{supply} - EMF_{back}}{R}$$



The variation of the back EMF by motor speed:

- When a DC electric motor is first started, the **coil is stationary** so the **back EMF = 0.** The net EMF of the coil is maximum.
- Once the armature begins to spin, the back EMF opposing the supply EMF is induced. The net EMF on the coil reduces
- As the **coil rotates faster**, the **back EMF increases** and the difference between the constant supply EMF and back EMF becomes smaller.
  - $EMF_{Supply} > EMF_{back}$. Torque is required to overcome mechanical resistance of axle.

The effect of back EMF on current and torque:

- When a DC electric motor is first started, the net EMF is high and so current flowing through the coil is **high** and equal to supply current
- As the coil rotates faster, the back EMF increases so total current flowing through the motor is reduced once the motor is in motion
- Due to mechanical resistances in the motor, $EMF_{Supply} > EMF_{back}$
- The increase in total current with decrease in speed makes a motor self-regulating to some extent

A motor at start-up has no back EMF. It can draw too much current and overheat and burnout.

- To protect the motor on start-up, an adjustable starting resistor is placed in the circuit.
- The resistor is removed when the motor's speed increases.

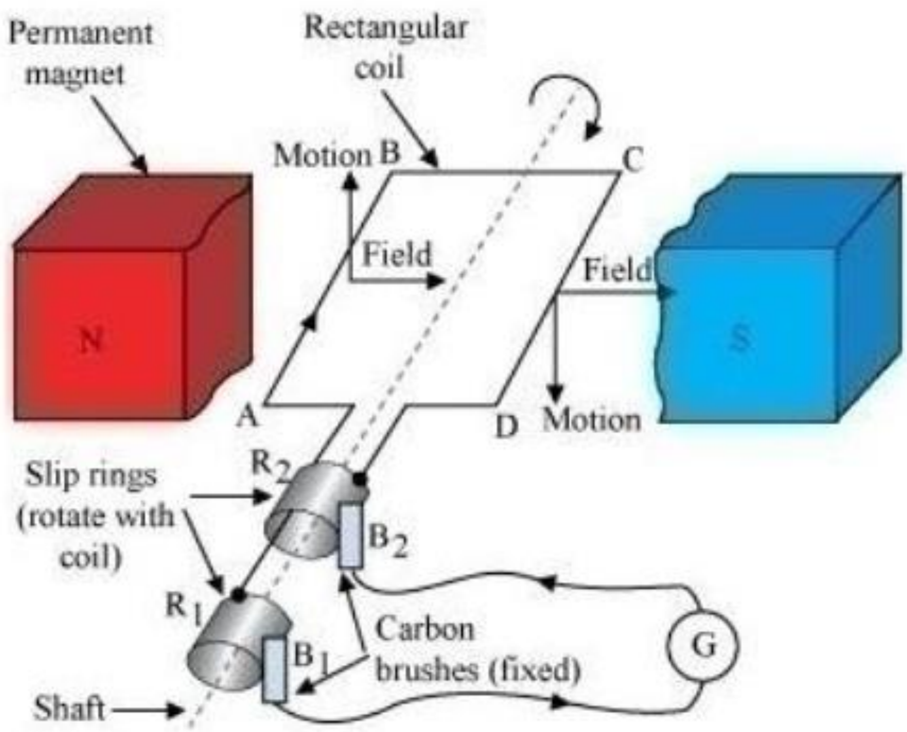
GENERATORS

OUTCOMES

- analyse the operation of simple DC and AC generators and AC induction motors (ACSPH110)

AC ELECTRIC GENERATOR

An AC generator is a device that converts mechanical energy to electrical energy in the form of **alternating current**.



The main components of an AC electric generator are:

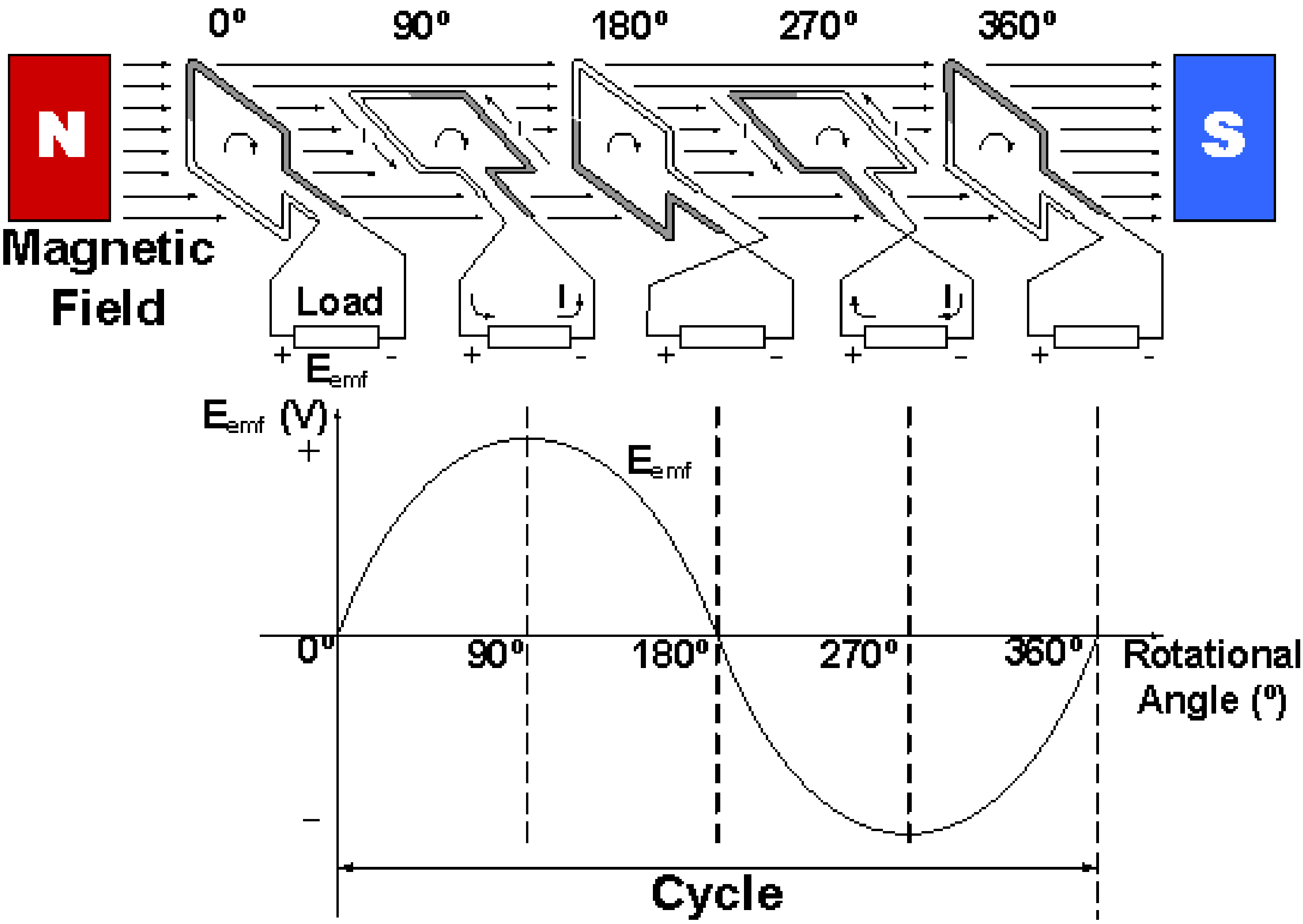
1) A pair of permanent magnets of pairs of electromagnets
2) Armature
3) Rotor coils
4) **Slip rings**
5) Brushes
6) Axle or shaft

Slip rings:

- Slip rings consist of two circular rings of metal mounted on the axle of the end of the armature
- Each end the coil is connected to one of the two slip rings on the axle. Brushes press against the slop rings providing a sliding electrical contact. The slip rings remain in contact with the same brush throughout the rotation in an AC generator.

A handheld generator uses a hand crank to provide **external torque** to **rotate a coil in a magnetic field**. The hand generator is more difficult to rotate when it is connected to a lightbulb compared ton when it is disconnected.

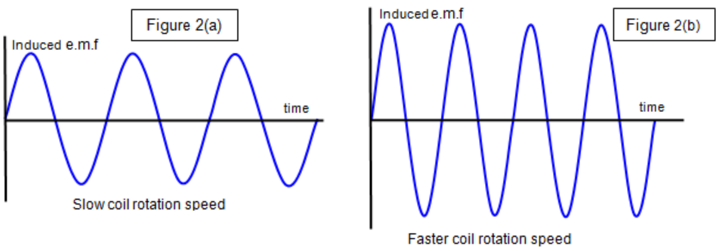
The lightbulb (closes the circuit) connects the coil ends in a closed circuit. The induced EMF therefore drives a current, which opposes the original change in magnetic flux (Lenz's law) by creating magnetic poles on which the external magnetic field exerts retarding force.

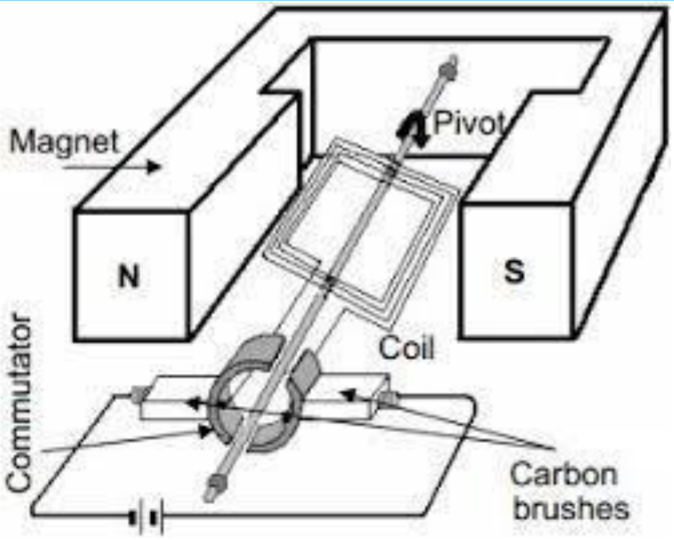


FREQUENCY OF AC GENEARTORS

The **frequency of rotation** of an AC generator affects both the **frequency** and **amplitude** of the **induced EMF** and **current**.

- In a generator, a coil rotates in a constant magnetic field
- $\epsilon = -\frac{N\Delta\phi}{\Delta t}$
    - With faster rotation, the flux through the coil changes at a faster rate. Therefore induced EMF will be greater.



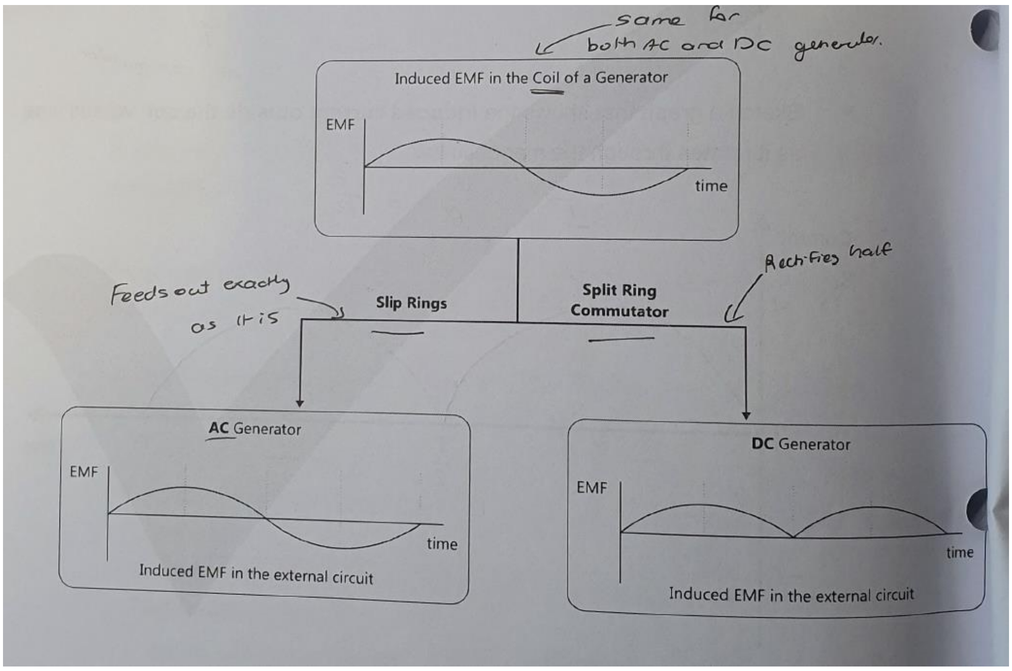


The main components of a DC generator are:

1) A pair of permanent magnets or pairs of electromagnetic coils
2) An armature
3) Rotor coils
4) **A split ring commutator**
5) Brushes
6) An axle or shaft

A split ring commutator:

- The commutator is a broad ring of metal mounted on the axle at the end of the armature and cut into an even number of separate bars. Each opposite pair of bars is connected to one coil.
- The commutator provides points of contact between the rotor coils and the external electric circuit. It serves to reverse the direction of current flow in the **external circuit** once every half-turn of the generator. This ensures that the current is always in the same direction (DC)



REGENERATIVE BREAKING

Regenerative breaking is used in vehicles that contain electric motors. The braking action arises when the motor is instead used as a generator.

- The generator **converts the vehicles kinetic energy into electrical energy**
- The vehicle experiences a **braking force due to Lenz's Law.**

The following flowchart:

1) Vehicle's wheel rotates and turn the coil in the motor
2) The coil in the motor experiences a change in flux. There is an induced EMF and current
3) The motor now acts as a generator
4) The coil experiences an opposing force due to Lenz's law.
5) The force opposes the rotation of the coil and the wheels and provide a braking force.
6) The kinetic energy of the vehicle is converted to electrical energy and stored.

Regenerative braking is similar to **eddy current braking** as both rely on induction, however there are important differences:

- In regenerative braking, **current is induced in the coil** of the motor/generator, not the metal wheel
- In regenerative braking the **electrical energy is stored** not dissipated as heat.

The purpose of regenerative breaking is to save some of the kinetic energy of the vehicle as electrical energy and then re-use it to power the vehicle. This makes the vehicle more efficient.

TRANSFORMERS AND TRANSMISSION

OUTCOMES

- analyse quantitatively the operation of ideal transformers through the application of: (ACSPH110)
  - $\frac{V_p}{V_s} = \frac{N_p}{N_s}$
  - $V_p I_p = V_s I_s$
- evaluate qualitatively the limitations of the ideal transformer model and the strategies used to improve transformer efficiency, including but not limited to:
  - incomplete flux linkage
  - resistive heat production and eddy currents
- analyse applications of step-up and step-down transformers, including but not limited to:
  - the distribution of energy using high-voltage transmission lines

POWER LOSS DURING TRANSMISSION

Power is lost in the transmission lines due to heating. The amount of power lost is given by:

$$P_{loss} = I^2 R = I V_{drop} = \frac{V_{drop}^2}{R}$$

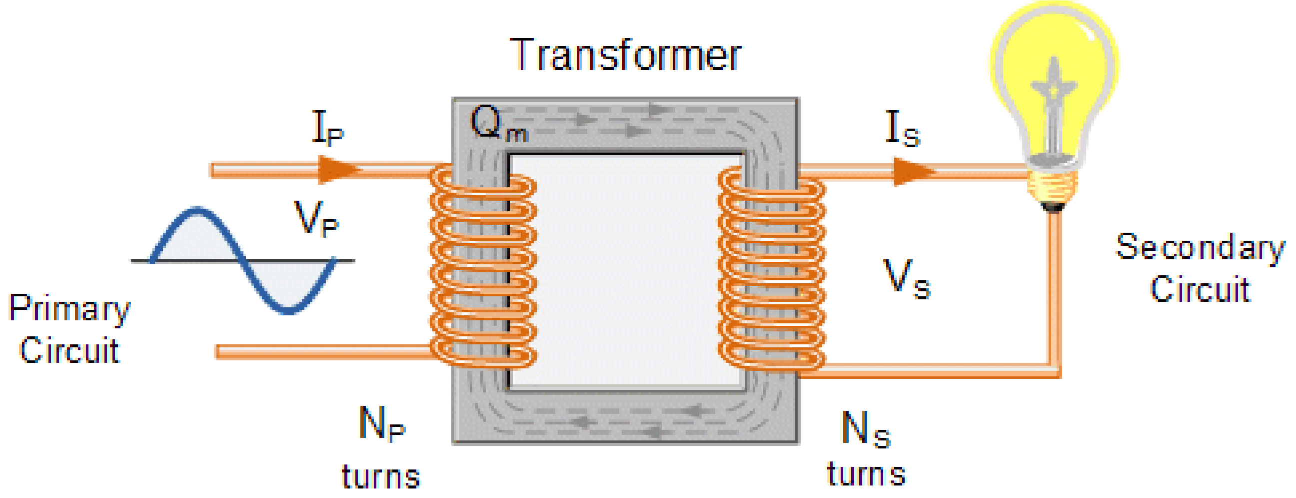
When transmitting power, it is economically beneficial to use **high voltage** and **low currents**.

THE TRANSFORMER

A transformer is a device that uses **electromagnetic induction** to transfer energy from one AC system to another. A transformer can accept energy at one voltage and deliver it at another voltage.

A transformer is used to alter the voltage in AC circuits.

- It consists of two coils of wire around the same laminated soft iron core.
- The **primary coil** is the **input coil** of the transformer, the **secondary coil** is the **output coil**.



OPERATION OF A TRANSFORMER

The operation of a transformer is based on the principle of electromagnetic induction. A **changing current in one coil** causes an **induced EMF** in the other coil.

- An **alternating current** in the **primary coil** produces an **oscillating magnetic flux**.
- The **secondary coil** experiences the changing magnetic flux and hence an EMF and **alternating current** are **induced** in the **secondary coil** by Faraday's Law.
- The AC current induced in the secondary coil is out of phase with the AC current in the primary coil, but has the same **frequency** as the current in the primary coil.

FEATURES OF A TRANSFORMER

WINDINGS

- The coil which is connected to the **input AC voltage source** is called the **primary winding.** (N_p)
- The coil which is connected to a **load** is called the **secondary winding**. (N_s)

AC SUPPLY

- An AC supply is connected to the primary coil and provides an alternating current in the primary coil.
- The AC passing through the primary coil produces continuously changing magnetic flux.

COMMON IRON CORE

- The primary purpose of the common iron core is to increase the strength of the magnetic field, and to provide a medium in which nearly all the magnetic flux produced by the primary coil is **trapped** and **passed** through he secondary coil, **minimising the flux leakage** and improving the efficiency of the transformer.

RELATIONSHIPS BETWEEN PRIMARY AND SECONDARY COILS

$$\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$$

- If $N_p > N_s$, then $V_p > V_s$. It is a **step down** transformer
- If $N_p < N_s$, then $V_p < V_s$. It is a **step up** transformer

IDEAL TRANSFORMERS

An **ideal transformer** with a resistive load is one in which **energy losses** in the transformer windings and core can be **neglected**. The power input = power output.

EFFICIENCY OF TRANSOFRMERS

A real transformer is not ideal, and some power is wasted. However, a transformer does not have moving parts so there is no power loss from friction. A transformer is thus very efficient with an efficiency of 99%.

$$Efficiency = \frac{Energy\ output}{energy\ input}$$

IMPROVING THE EFFIEINCY OF A TRANSFORMER

Name of Power Loss	Detail	Mitigation
Flux leakage	Not all the magnetic flux links both the primary and secondary coil, reducing the efficiency of the transformer.	Use ferromagnetic core, reduce air gaps, reduce distance between coils
Resistance in coils (Joule heating)	The wires have finite resistance and so will heat up, further increasing resistance and power loss	Use low resistance wire (increase thickness) and use cooling strategies
Eddy currents in core	Any currents in the core cause heating and energy loss due to electrical resistance, and can opposite the existing magnetic field in the core	Laminate ferromagnetic core to limit electrical properties while maintain magnetic properties.
Hysteresis	The ferromagnetic core magnetises semi-permanently, so it takes energy to reverse the existing magnetisation energy time the input AC reverses	Use 'soft' magnetic materials which do not consume a lot of energy when the magnetisation is reversed.

ROLE OF TRANSFORMERS

It is more efficient to transmit power over long distances at high voltage and low current, in order to reduce power loss in transmission lines.

- The role of step-up transformers at an electrical power station is to increase the voltage to reduce power loss in the transmission lines.
- The role of transformers in electricity substations is to progressively reduce the voltage as it comes closer to the consumer.

SUMMARY EQUATION SHEET

CHARGES

$$F = qE$$
$$E = \frac{V}{d} = \frac{F}{q}$$
$$a = \frac{F}{m} = \frac{qE}{m}$$

$$\Delta K = W = qEd = qV$$
$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$
$$V = \frac{W}{q}$$

PROJECTILE MOTION WITH CHARGES

$$v_\perp = u_\perp$$
$$\Delta s_\perp = u_\perp t$$

$$v_\parallel{}^2 = u_\parallel{}^2 + 2a_\parallel \Delta s_\parallel$$
$$\Delta s_\parallel = u_\parallel t + \frac{1}{2}u_\parallel{}^2$$
$$v_\parallel = u_\parallel + a_\parallel t$$

MAGNETIC FIELD & FORCES

$$B = \frac{\mu_0 I}{2\pi r}$$
$$F = qv_\perp B = qvB\sin\theta$$

$$B = \frac{\mu_0 NI}{L}$$

MOTOR EFFECT

$$F = BI_\perp L = BIL\sin\theta$$

$$\tau = r_\perp F = rFsin\theta$$

$$\frac{F}{L} = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{d}$$

ELECTROMAGNETIC INDUCTION

$$\phi = B_\parallel A = BAcos\theta$$
$$\epsilon = -\frac{N\Delta\phi}{\Delta t} = -\frac{N\Delta BAcos\theta}{\Delta t}$$
$$P_{loss} = I^2 R = IV_{drop} = \frac{V_{drop}^2}{R}$$

$$EMF_{coil} = EMF_{supply} - EMF_{back} = I_{coil}R$$
$$I = \frac{V}{R} = \frac{V_{supply} - EMF_{back}}{R}$$